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Assignment - Cyber Security - Virtualization and Cloud Basics

Module 2 – Fundamental of Operating Systems & Networks

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1. Difference Between Hardware and Software

Answer -

Hardware refers to the physical components of a computer that can be seen and touched. Examples include the CPU, keyboard, monitor, mouse, hard disk, and RAM.

Software refers to a set of programs and instructions that tell the hardware how to perform specific tasks. Software cannot function without hardware.

Hardware	Software
Physical components of a computer	Programs and applications
Can be touched and seen	Cannot be touched
Examples: CPU, RAM, Keyboard	Examples: Windows, Linux, MS Word
May wear out over time	Can be updated or upgraded

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2. Define IP Address Range and Private Address Range

Answer -

An IP (Internet Protocol) address is a unique number assigned to a device connected to a network. It helps identify and communicate with other devices.

Private IP Address Ranges

Private IP addresses are used within local networks and cannot be accessed directly from the internet.

- Class A: 10.0.0.0 – 10.255.255.255
- Class B: 172.16.0.0 – 172.31.255.255
- Class C: 192.168.0.0 – 192.168.255.255

Private IP addresses are commonly used in homes, offices, and cloud environments.

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3. Explain Network Protocol and Port Number

Answer -

Network Protocol

A network protocol is a set of rules that allows devices to communicate over a network.

Common protocols include:

- HTTP – Used for web browsing
- HTTPS – Secure web communication
- FTP – File transfer
- SMTP – Sending emails
- TCP/IP – Internet communication

Port Number

A port number is a logical communication endpoint used by applications and services.

Examples:

Service	Port Number
HTTP	80
HTTPS	443
FTP	21
SSH	22
SMTP	25

Port numbers help identify specific services running on a device.

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4. Explain Types of Network Devices

Answer -

Router

Connects different networks and forwards data between them.

Switch

Connects devices within the same network and transfers data efficiently.

Hub

Connects multiple devices and broadcasts data to all connected devices.

Firewall

Monitors and controls incoming and outgoing network traffic.

Access Point

Provides wireless connectivity to devices.

Modem

Connects a local network to the internet service provider.

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5. Which Tools are Used for Data Backup and Recovery?

Answer -

Data backup and recovery tools are used to protect important data and restore it in case of loss, corruption, or system failure.

Common backup and recovery tools include:

- Windows Backup and Restore
- Acronis True Image
- Veeam Backup & Replication
- EaseUS Data Recovery
- Google Drive Backup
- OneDrive Backup

Benefits:

- Prevents data loss
- Supports disaster recovery
- Improves business continuity

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6. Explain HTTP and HTTPS Protocols

Answer -

HTTP (HyperText Transfer Protocol)

HTTP is a communication protocol used to transfer web pages between a web browser and a web server.

Features:

- Fast communication
- Data is transmitted in plain text
- Less secure

Default Port: 80

HTTPS (HyperText Transfer Protocol Secure)

HTTPS is the secure version of HTTP. It uses encryption to protect data exchanged between users and websites.

Features:

- Secure communication
- Protects sensitive information
- Uses SSL/TLS certificates

Default Port: 443

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7. What is SSL and TLS Security?

Answer -

SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are security protocols used to protect data transmitted over a network.

Their main purpose is to ensure that information exchanged between a client and a server remains private and secure.

Functions:

- Encrypt data
- Protect against data theft
- Verify website authenticity
- Secure online transactions

TLS is the modern and more secure version of SSL.

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8. Explain the MAC Address

Answer -

MAC (Media Access Control) Address is a unique hardware address assigned to a network interface card (NIC) by the manufacturer.

A MAC address is used to identify devices within a local network.

Example:
00:1A:2B:3C:4D:5E

Characteristics:

- Unique for every device
- Used for local network communication
- Works at the Data Link Layer of the OSI model

Module 5 – Cryptography and Network Security

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1. Explain Mitigation in Reference to Cyber Security

Answer -

Mitigation in cyber security refers to the actions taken to reduce or eliminate security risks and threats.

The goal of mitigation is to minimize the impact of cyber attacks and protect systems from damage.

Examples:

- Installing firewalls
- Using antivirus software
- Applying security patches
- Conducting regular backups
- Implementing access controls

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2. What are the Differences Between IDS and IPS?

Answer -

IDS (Intrusion Detection System)	IPS (Intrusion Prevention System)
Detects suspicious activities	Detects and blocks attacks
Generates alerts	Takes preventive action
Passive monitoring system	Active protection system
Does not stop attacks	Can automatically stop attacks

Both systems help improve network security and threat detection.

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3. Explain Network-Based IDS

Answer -

A Network-Based Intrusion Detection System (NIDS) monitors network traffic to identify suspicious activities and potential security threats.

It analyzes packets moving across the network and generates alerts when unusual behavior is detected.

Advantages:

- Continuous monitoring
- Early threat detection
- Improved network visibility

Examples:

- Suricata
- Snort

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4. Explain How SSL and TLS Work

Answer -

SSL and TLS establish a secure connection between a client and a server.

Working Process:

1. The client requests a secure connection.
2. The server sends its digital certificate.
3. The client verifies the certificate.
4. Encryption keys are exchanged securely.
5. Data is encrypted before transmission.
6. Secure communication begins.

This process protects data from interception and unauthorized access.

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5. What is Symmetric Key Cryptography and Asymmetric Key Cryptography?

Answer -

Symmetric Key Cryptography

In symmetric encryption, the same key is used for both encryption and decryption.

Examples:

- AES
- DES

Advantages:

- Fast
- Efficient for large amounts of data

Asymmetric Key Cryptography

In asymmetric encryption, two keys are used:

- Public Key
- Private Key

Examples:

- RSA
- ECC

Advantages:

- Higher security

- Secure key exchange

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6. Explain How to Secure Servers and Personal Computers

Answer -

Servers and personal computers can be secured by following good security practices:

- Install antivirus and anti-malware software
- Keep operating systems updated
- Use strong passwords
- Enable multi-factor authentication
- Configure firewalls
- Perform regular backups
- Limit user permissions
- Monitor system activity

These measures help protect systems from cyber threats and unauthorized access.

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7. Explain Suricata and SolarWinds

Answer -

Suricata

Suricata is an open-source network security tool used as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS).

Features:

- Real-time traffic monitoring

- Threat detection
- Malware detection
- Network security analysis

Suricata helps organizations identify and respond to security threats quickly.

SolarWinds

SolarWinds is a network and system monitoring platform used to manage IT infrastructure.

Features:

- Network performance monitoring
- Server monitoring
- Device management
- Fault detection and reporting

SolarWinds helps administrators monitor network health and maintain system performance.

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8. Describe VPN and IPSec

Answer -

VPN (Virtual Private Network)

A VPN creates a secure and encrypted connection between a user and a network over the internet.

Benefits:

- Protects privacy
- Secures remote access

- Encrypts internet traffic

IPSec (Internet Protocol Security)

IPSec is a set of security protocols used to secure IP communications by encrypting and authenticating data packets.

Features:

- Data encryption
- Authentication
- Data integrity
- Secure communication over networks

IPSec is commonly used to implement secure VPN connections.